Projet — proposal

Nov 4, 2026 · @Joost Van kerkhoven

Purpose

Projet is a playable sketch timeline for planning a piece before any material exists. It sits at the **start** of the chain: you lay out the conceptual skeleton of a piece in Projet, then fill it with material made in Trajet, Effet and Goblet, and finally assemble everything in Reaper.

It combines two existing ideas:

- **Reaper's empty items**: post-its on a timeline marking conceptual key moments, which later become subprojects nested inside each other. That way of working stays; Projet removes the limit of a few lines of static text.
- **OpenMusic's maquette**: boxes placed in time, and boxes that contain their own timeline. Only the spatial metaphor is borrowed. There is no visual programming and no box computes anything (for now: see the future codebox under Box types).

The nested box structure in Projet is a rough blueprint of the nested Reaper project the piece will become. Part of Projet's design is discovering, while building and using it, which box types are actually worth having.

Core concepts

A piece is one timeline of tracks; tracks hold boxes; a box can hold its own timeline of tracks and boxes, to any depth.

Concept	What it is
Piece	The top-level Projet document: one root timeline.
Track	A horizontal lane holding boxes, as in Reaper. Boxes on one track may overlap (see Box types: Overlap).
Track folder	A group of tracks that can be folded into one row and unfolded again.
Box	An item on a track with a start, a length, a name, a colour, a type and a notes field.
Nested timeline	The optional timeline inside a box, with its own tracks and boxes. A box with one is a container box; it maps onto a Reaper subproject.

Concept	What it is
Depth level	How many containers a box sits inside. The piece is level 0. Always visible, so you never lose track of where you are.

Time model

Every container box has a switch that decides what its inner timeline does when you resize the box; the ruler can show seconds or proportions.

Window or stretch, per box

- **Window** (Reaper-like): the inner timeline has its own fixed length. Resizing the box shows more or less of it, like trimming a subproject item.
- **Stretch** (maquette-like): the inner timeline scales with the box. Everything inside becomes proportionally faster or slower, sounds included.
- Typical use: sketch in stretch while proportions are still moving, switch to window once durations become real. New container boxes default to stretch, since sketching starts in proportions.

Ruler: seconds or proportions

- The ruler of any timeline can show minutes:seconds or proportions of the parent box (0–1, or percent).
- Snap grid in fractions: halves, thirds, quarters, fifths, golden section. You can sketch "the turn happens two-thirds of the way in" and commit to seconds later.
- The piece itself needs a working total length so proportions mean something; it can change at any time.

Box types and extensibility

Every box has the same shell and one swappable content type; the notes field belongs to the shell, so it survives every type change.

The shell (common to all boxes)

- Start, length, name, colour, track
- Notes field: free text for the conceptual idea at that moment, always present
- Optional nested timeline (makes it a container)
- Window/stretch switch

- Type, switchable at any time without losing the notes; takes (A/B/C…), one of them active

Content types

Type	Output	Status
Text	None: notes and memory aid only	First version
Sound	Plays a WAV file copied into the project folder: region, gain, fades, crossfades; no varispeed	First version
Container	Plays whatever its nested timeline holds	First version
Codebox	Generates musical structure from code in the user's own composition language	Future, once that research matures
Others	To be discovered while using Projet	Open
Board	A mini whiteboard for working out internal logic (cards, arrows, drawing, images, sound cards); silent in the piece unless a sound card is marked as temporal out	Later stage, after timeline, sound boxes and takes

Built to be extended

- Each content type is a separate class behind one common interface: draw itself, edit itself, save and load itself, and optionally produce audio (or later MIDI or events) during playback.
- Adding a type means writing one new class and registering it; the timeline, views and file format stay untouched.
- Files store each box's type by name with its own data block, so a file made later with new types still opens in an older version (unknown boxes show as text boxes with their notes).

Notes editor

- Rich text, not a plain line: bold, italic, headings, bulleted lists and clickable links (for example a URL to an inspirational reference). Links open in the browser.
- The inspector shows a generous editor; an Expand button opens it in a large separate window for longer writing.
- The text box type uses the same editor, so a text box can hold a full page of thinking.

Takes

- A box can hold several takes (A, B, C…). Exactly one is active, and playback and export use the active one.
- The box shell is shared across takes: position, name, colour and the main notes. Each take has its own content (its nested timeline, sound or text) and an optional one-line take note.
- The box takes the length of its active take. Duplicate take copies the active take as a starting point for a variation. Switching takes is one click on the box or in the inspector.
- Works for every box type: alternative sound files, alternative inner structures of a section, alternative wordings of an idea.

Overlap

- Boxes on one track may overlap, so sound sections can blend into each other.
- Each box has an opacity. Overlapping boxes are drawn see-through, and the overlap region is marked as a crossfade.
- Playback mixes overlapping sound boxes, with crossfade curves over the overlap. Opacity is visual only; the level is set by the box's gain.

Connections between boxes

Four mechanisms make logical and causal connections visible. Relations and materials go into the data model from the start; the board box is built as a later stage.

Mechanism	What it is	Where you see it	Stage
Relations	An arrow from one box to another, across tracks and levels, with a kind (prepares, answers, transforms into, contrasts with, or your own) and a short note	Curves above the timeline, toggled on and off; listed in the inspector; drawn across bands in strata	Data model from the start; drawing early
Materials	Tags such as "drone material" or "glissando idea", each with a colour	A colour stripe on each box; outline and timeline filter by material	Early
Origin	A box or take remembers which box or take it was made from	Inspector: "made from Suspension, take A"	Early, almost free

Mechanism	What it is	Where you see it	Stage
Board box	A box type holding a free 2D canvas for working out the logic inside a section	Thumbnail on the timeline; opens in its own editor window	Later stage

Board box contents

- Text cards, using the same rich-text editor as the notes
- Arrows between cards, with an optional label
- Simple shapes and freehand drawing
- Dropped images: sketches, score excerpts, screenshots
- Sound cards that play a clip when clicked; drag one onto the timeline to make it a sound box, or mark it as temporal out (below)
- Timeline links: a card pointing to a box (and take) in the piece; clicking it jumps there

Temporal out (as in OpenMusic's maquette). A sound card on a board can be marked as a temporal out. The board box then sounds when the timeline plays:

- Each temporal-out card has a start offset inside the board box (default 0:00), its own gain and fades, and is cut off at the box's end.
- Several temporal-out cards in one board mix together; cards that are not marked stay silent and only play when clicked.
- On the board, a temporal-out card carries an "out" mark; on the timeline, the board box shows a small waveform strip of what it plays, so a sounding board is told apart from a silent one at a glance.
- In playback the cards are flattened into the same clip list as sound boxes, so timing is just as sample-accurate.
- A temporal out never stretches. For a stretched version, import an externally stretched copy of the sound and mark that card as the temporal out.

Mockups 4 and 5 (second row of the canvas) show relations and materials on the timeline, and the board box editor. Relations are made by ⌥-dragging from one box onto another (Control stays free for secondary click).

Composition framework

Projet speaks the user's own framework: the ten energy models (Vande Gorne) and four semiotic levels (Peirce via Bayle: I-son, Di-son, Mé-son, plus Thoresen's metonymic sign). All four layers below go into the first version; layer 4 is a first pass to refine with use.

Layer	What it adds	Where it shows
1. Structured fields	Main energy + optional second energy; semiotic level; energy family derived automatically (matérielles, gestion globale, gestuelles, mouvements, lois)	Inspector, outline, filters
2. Visual language	Energy → a morphological shape drawn along the box; semiotic level → fill (I-son solid / Di-son hatched / metonymic dotted / Mé-son outline), changeable along the box; colour mode switchable (family, level, material, own)	Every box, every view
3. Structure	Semiotic track template (I-son, Di-son, Mé-son folders); overview lanes for energy distribution and semiotic profile	Timeline, strata
4. Active parameters (v1)	Each energy's variation factors as drawable curves that reshape its glyph; semiotic relation kinds (develops, metaphor of); energy shapes drawn as gestures on boards	Box editor, boards, relations

Energy models and their variation factors (from the source text)

Energy	Family	Variation factors
Percussion-résonance	Matérielle	Dynamique, halo harmonique, attaque, entretien
Frottement-granulation	Matérielle, gestuelle	Vitesse, grain, entretien itératif
Accumulation de corpuscules	Gestion globale (mobile)	Grain, masse, calibre, densité
Oscillation	Loi	Force de déclenchement, vitesse, densité, fréquence, spectre, espace G-D, profondeur
Balancement	Gestuelle	Vitesse, densité, fréquence, spectre, espace G-D, profondeur
Rebond	Gestuelle, loi	Vitesse, halo harmonique, matière

Energy	Family	Variation factors
Flux	Gestion globale (immobile), mouvement	Densité, profil de masse
Pression-déformation	Mouvement	Vitesse, profil de masse or profil mélodique, enveloppe spectrale
Rotation	Mouvement	Vitesse, fréquence, espace
Spirale	Mouvement	As rotation, plus direction of acceleration (toward high or low)

Framework file. Energies, families, factors, glyph rules and semiotic levels live in an editable framework file inside Projet, not hard-coded. Refining the framework, or adding another (Schaeffer′s morphological criteria, Thoresen′s aural sonology signs), then needs no new code.

Boards as the deep home of the framework. A board can contain a morphogram region: time across, register (low to high) up. Drawing a gesture with an energy tool renders that energy′s shape along the drawn path, with stroke weight as dynamics and factor curves editable afterwards. Cards carry a semiotic level and use the same fills. This is the Acousmographe idea turned around: drawing first, sound later.

From Thoresen and Schaeffer (decided: all built, in the stages listed)

Thoresen′s three levels of organisation map onto Projet′s own structure: level 1 (sound-object) belongs on boards and morphograms, level 2 (compound sound-patterns) is the box, level 3 (form-building patterns) is the container, its tracks and its junctions.

Tool (source)	In Projet	Stage
Time-field positioning (Thoresen 6.3): separate, bridged, joint, close, hinged, overlapping, superimposed	A junction between consecutive boxes, drawn as a small sign at the seam; mostly derived from the gap or overlap, adjustable	4
Time-field demarcation (Thoresen 6.3): vague, open, conclusive, cut, disjointed, deferred	How a box ends, chosen by the composer, drawn at the box′s end	4
Layer functions (Thoresen 7.2): foreground, middleground, background	A box property that sets its visual weight	4
Dynamic form (Thoresen 8): forward-, presence-, backward-oriented, neutral	Wedges on containers in an overview lane: the energy direction of the large form	4

Tool (source)	In Projet	Stage
Form-building transformations (Thoresen 9.3): simple/complex, part/whole, one/many, distinctive/anonymous	Relation kinds between boxes, visible in strata	8
Typology and the seven morphological criteria (Schaeffer: mass, harmonic timbre, dynamic, grain, allure, melodic profile, mass profile)	A second descriptor set in the framework file, used mainly on boards, so boxes are not overloaded	9, 11

The four semiotic levels. Ordered from concrete to abstract, which sets the fills and the semiotic-profile lane: I-son (iconic: the source as is), Di-son (diagram: the source as model, thread kept), metonymic (association by contiguity: a part or a neighbour stands for the whole, Thoresen's addition to Peirce), Mé-son (metaphor: the source hidden or arbitrary). The levels live in the framework file, so names, order and fills can be changed without code.

Views

All views show the same data; you switch between them, and the outline panel is always present. A 3D view was considered and dropped.

View	What it does	Best for
Tracks + folders	Reaper-like editing view; track folders fold and unfold	Day-to-day editing
Dive-in	Double-click a container: its inner timeline fills the window; a context strip shows the parent with your box highlighted; breadcrumb trail on top	Working inside one level
Inline expansion	Selecting a container magnifies it: its span stays highlighted across the piece's tracks, and its inner tracks open full-width beneath, with their own ruler	Seeing a detail against its surroundings
Strata	All depth levels at once as stacked bands, each box at its true proportional position under its parent	Overview of the whole nested structure; preview of the Reaper subproject tree

View	What it does	Best for
Outline panel	Side panel with the box hierarchy, depth level per row, current position highlighted	Never losing track of depth levels
3D (later)	Dropped on 2026-11-04	Exploration, once the 2D views are in use

Mockups of dive-in, inline expansion and strata: all three show one example piece (Étude II, 12:00, three levels deep), with the outline panel on the left and the box inspector (type, window/stretch, notes) on the right.

Technical scope

Projet is macOS-only, stereo, plain playback, with a linear time base and its own project folder.

Area	Decision
Platform	macOS on Apple Silicon, JUCE, code copied from Trajet and Effet (no shared library); repo Joostiepoost/Projet
Project storage	One folder per piece. Every imported sound is converted into a compact placeholder copy at the project sample rate (proposed: Ogg Vorbis, high quality; FLAC as an option). The path of the original is remembered, so it can later be swapped for the production-quality file. Importing the same file twice reuses the copy
Audio	Stereo only. No plugins. Per sound box: region, gain, fade in/out; crossfades where boxes overlap. No varispeed
Time base	Linear: minutes and seconds, or proportions of the parent. No bars and beats
Opacity	Visual only
Takes	The box takes the length of its active take
New containers	Stretch by default

Proposed project folder:

```
Étude II/
  Étude II.projet    the piece: tracks, boxes, takes, relations, notes (text)
  Sounds/             copies of every sound used
  Boards/             images placed on boards
  Exports/            Reaper exports
  Backups/            automatic snapshots
```

Build stages

Each stage ends with something you can test, as with Trajet and Effet; the first stage tests playback timing in depth before anything is built on it.

1. **Engine and project folder.** App shell, create/open/save a project folder, Vorbis/FLAC import, tracks, sound boxes on a flat timeline, transport, playback with gain and fades. Timing tests.
2. **Editing.** Text boxes, the rich-text notes editor (large-window mode, links), type switching, move/resize/split/copy, overlap with opacity and crossfades, undo.
3. **Rulers and tracks.** Seconds and proportion rulers, fraction snapping, track folders.
4. **Framework, layers 1–3.** Framework file, energy and semiotic fields, energy shapes and semiotic fills on boxes, colour modes, semiotic track template, overview lanes.
5. **Nesting.** Container boxes, window/stretch, nested playback, dive-in with context strip and breadcrumbs, outline panel.
6. **Magnified inline expansion.**
7. **Takes.** Per box, with take notes and duplicate take.
8. **Relations, materials, origin.** Option-drag to relate, curves on the timeline, material stripes and filters, semiotic relation kinds.
9. **Framework, layer 4 (v1).** Factor curves per energy reshaping the glyphs.
10. **Strata view.**
11. **Board box.** Cards, arrows, shapes, pen, images, sound cards, timeline links, morphogram regions with energy gesture tools.
12. **Reaper export.**

Later: links to Trajet, Effet and Goblet output; the codebox.

Export to Reaper

The format is decided later in the build; the data model only has to keep the mapping simple.

- Container box → Reaper subproject (its own .rpp), nested to the same depth.

- Track and track folder → Reaper track and folder track.
- Text box → Reaper empty item with the notes as item text.
- Sound box → media item on the same track and position.
- Reaper's project format is plain text, so Projet can write .rpp files directly. Whether the export is a one-off template or something you repeat as the sketch changes is still open.

Open questions

Everything needed to start building is decided (see the decisions log). These can be settled during the build:

- [ ] Sounds at a sample rate other than the project's: convert on import, or convert during playback?
- [ ] Linking boxes to Trajet, Effet and Goblet projects or renders
- [ ] Reaper export: exact mapping, and one-off template versus repeatable export
- [ ] Relation kinds: fixed list, your own list, or both
- [ ] Board box details (a later stage)
- [ ] Codebox (once the composition-language research is further along)
- [x] ~~Semiotic levels: Bayle's three only, or Thoresen's quaternity (adds the metonymic sign) as an option in the framework file?~~
- [x] ~~Which of Thoresen's level-3 tools go into stage 4 (junctions, demarcation, layer functions, dynamic-form wedges)?~~

Decisions log

Date	Decision
2026-11-04	Projet sits at the start of the chain: a playable sketch timeline, not a file manager at the end.
2026-11-04	Boxes sit on tracks; track groups can be folded.
2026-11-04	Window/stretch switch per container box; ruler in seconds or proportions.
2026-11-04	Every box keeps a notes field; the type is switchable without losing it.
2026-11-04	Box types are open-ended classes, leaving room for a future codebox.

Date	Decision
2026-11-04	2D first; 3D later as an optional view.
2026-11-04	Outline panel is part of the app from the start, to keep track of depth levels.
2026-11-04	Inline expansion magnifies the selected container full-width beneath the piece timeline.
2026-11-04	Notes and text boxes get a rich-text editor with links and a large-window mode.
2026-11-04	Boxes may overlap, drawn with opacity, mixed with crossfades.
2026-11-04	Boxes hold takes (A/B/C…), one active; built into the data model from the start.
2026-11-04	Keep both dive-in and magnified inline expansion; the magnified view needs no connecting funnel, just the highlighted span above and the inner tracks below.
2026-11-04	Relations between boxes and material tags go into the data model from the start; origin tracking too.
2026-11-04	Board box (mini whiteboard) planned as a later box type.
2026-11-04	Sound cards on a board can be dragged onto the timeline to become sound boxes.
2026-11-04	3D view dropped.
2026-11-04	Own project folder per piece; sounds copied in.
2026-11-04	Stereo only, no plugins; gain, fades, crossfades; no varispeed.
2026-11-04	Linear time base only; opacity visual only; box follows its active take's length; new containers default to stretch.
2026-11-04	Repo Joostiepoost/Projet created.
2026-11-05	Ogg Vorbis placeholders by default, FLAC as a per-project option.

Date	Decision
2026-11-05	Composition framework (energies, semiotic levels) built in, all four layers, layer 4 as a first version; deep integration in boards.
2026-11-05	Stage 1 pushed to Joostiepoost/Projet for testing.
2026-11-05	Four semiotic levels: Thoresen's metonymic sign added between Di-son and Mé-son (dotted fill).
2026-11-05	All Thoresen level-3 tools adopted: junctions, demarcation and layer functions and dynamic-form wedges in stage 4; transformations as relation kinds in stage 8; Schaeffer's criteria as a second descriptor set (stages 9 and 11).
2026-11-05	Board sound cards can be marked as temporal out, so a board box sounds when the timeline plays (stage 11).
2026-11-05	Temporal outs do not follow stretch; stretched versions are imported as separate sounds.